

Clustering

Part 2

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Agenda

1. Tweets as data
2. Words → topics
3. Application

Social Media

- Unprecedented access to our leaders
 - (If they let us)



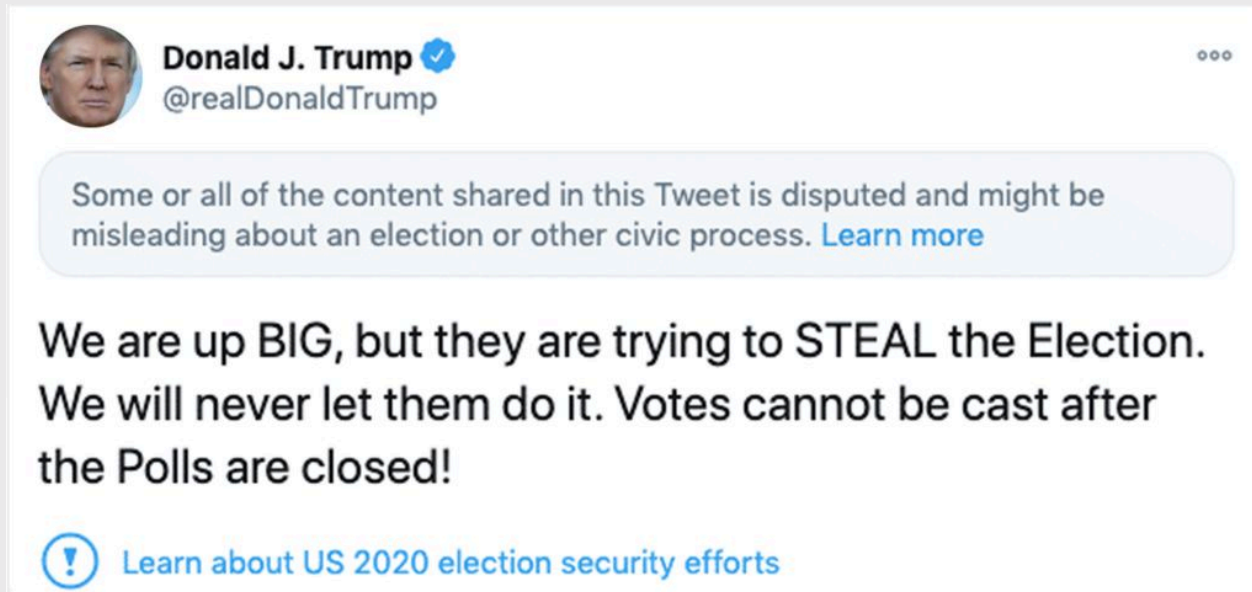
Social Media

- Unprecedented access to our leaders
 - (If they let us)



Social Media

- For researchers, social media is two things
 1. A source of **data**
 2. An object of **interest**

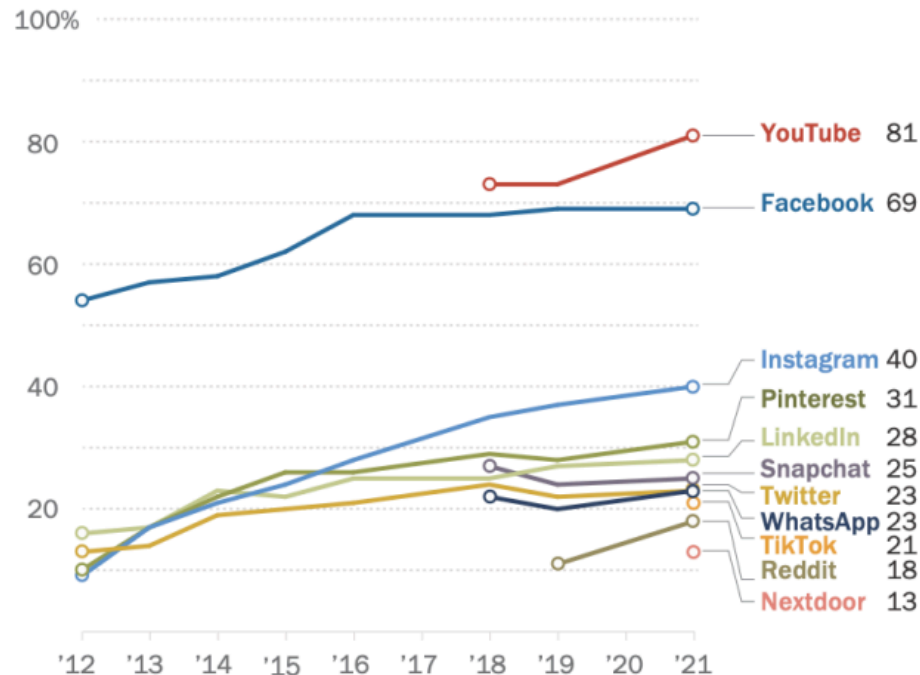


Twitter as Data

- Not the most popular social media app

Growing share of Americans say they use YouTube; Facebook remains one of the most widely used online platforms among U.S. adults

% of U.S. adults who say they ever use ...



Note: Respondents who did not give an answer are not shown. Pre-2018 telephone poll

Twitter as Data

- But an outsized platform for the elite
- As of 2020
 - every U.S. governor had a Twitter account
 - 49 had a Facebook account
 - 44 had an Instagram account
 - 44 had a YouTube account
- In professional networks, particularly media, Twitter is almost *lingua franca*

Twitter as Data

- Today?



Elon Musk  
@elonmusk · [Follow](#)



Twitter has had a massive drop in revenue, due to activist groups pressuring advertisers, even though nothing has changed with content moderation and we did everything we could to appease the activists.

Extremely messed up! They're trying to destroy free speech in America.



Readers added context



Multiple advertisers have confirmed they are pausing spending while they see what direction Elon Musk takes with the platform.

"The growing exodus of advertisers comes amid concerns Musk will scale back misinformation and security protections on the platform."

[theguardian.com/technology/202...](https://theguardian.com/technology/2022...)

Context is written by people who use X, and appears when rated helpful by others. [Find out more.](#)

Trump and Twitter

- Today, looking at Trump's tweets
 - Treating it as a **data source**
 - What can his tweets tell us about the man?

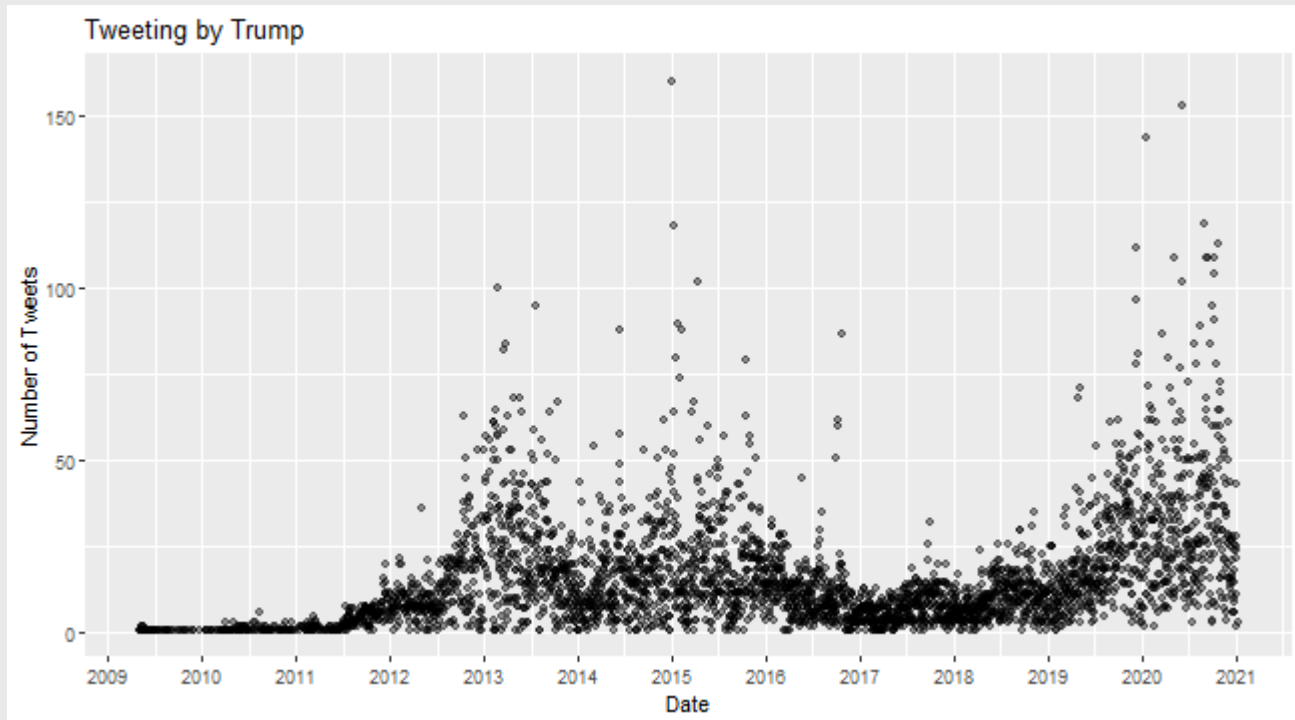
```
require(tidyverse)
tweets <-
read_rds(file="https://github.com/jbisbee1/DS1000_F2024/raw/main/data/T
```

- The **process**
 - Univariate visualisation: how often does he tweet?

```
p <- tweets %>%
  count(Tweeting.date) %>%
  ggplot() +
  geom_point(aes(x=Tweeting.date,y=n),alpha=.4) +
  scale_x_date(date_breaks = 'years',date_labels = '%Y') +
  labs(x="Date",y="Number of Tweets",title="Tweeting by Trump")
```

Trump and Twitter

p



Trump and Twitter

- Research questions abound!

1. What happened in...

- June of 2011?
- June of 2012?
- November of 2016? (duh)

2. Overtime increase during presidency?

3. Others?

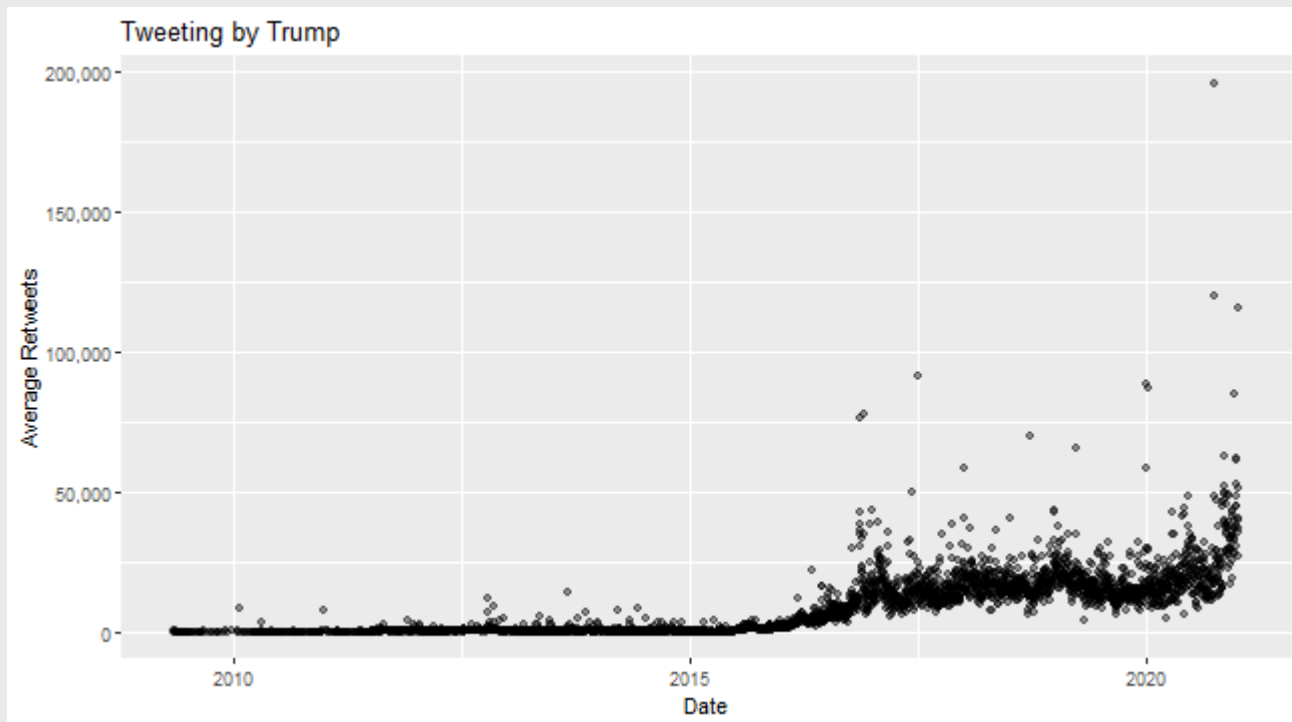
Trump and Twitter

- **Research Question:** Did Trump's twitter account benefit from his presidency?

```
require(scales)
p <- tweets %>%
  group_by(Tweeting.date) %>%
  summarize(AvgRetweet = mean(retweets)) %>%
  ggplot() +
  geom_point(aes(x=Tweeting.date,y=AvgRetweet),alpha=.4) +
  labs(x="Date",y="Average Retweets",title="Tweeting by Trump") +
  scale_y_continuous(label=comma)
```


Trump and Twitter

- **Research Question:** Did Trump's twitter account benefit from his presidency?
- **Yes**



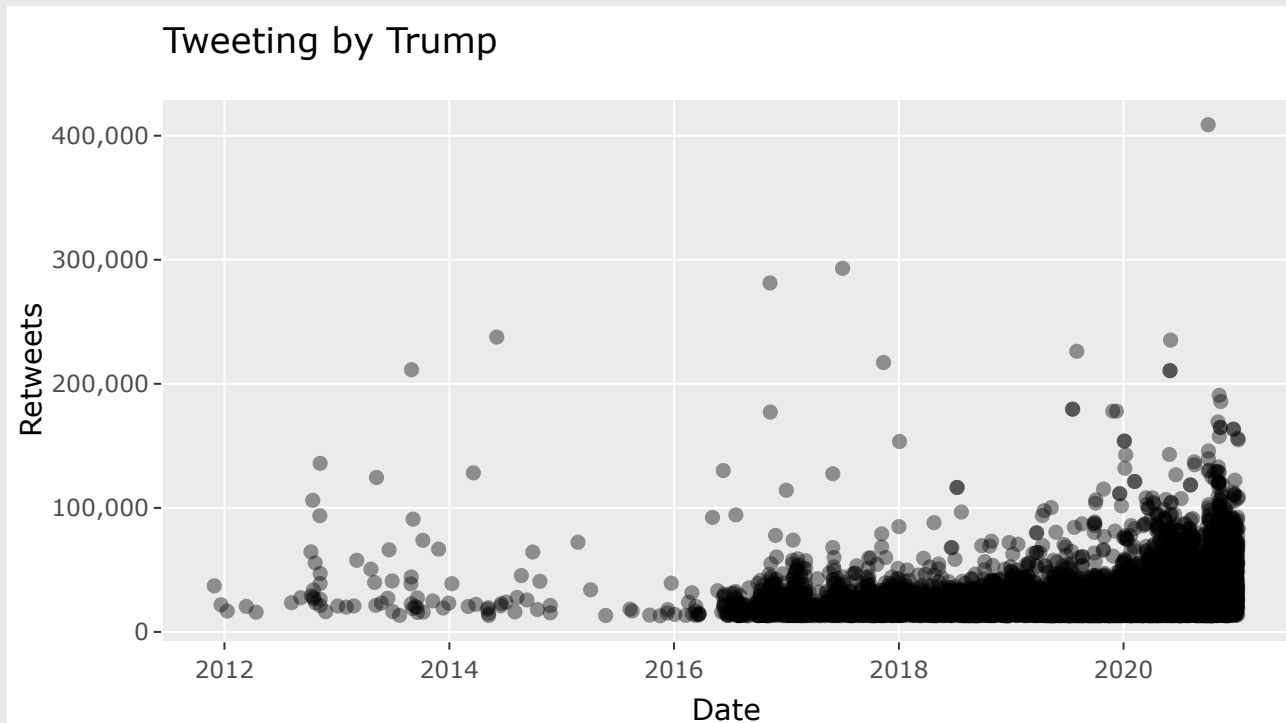
Looking with `plotly`

```
library(plotly)
gg <- tweets %>%
  filter(retweets > quantile(retweets,.75)) %>%

ggplot(aes(x=Tweeting.date,y=retweets,text=stringr::str_wrap(content,wi
= 60))) +
  geom_point(alpha=.4) +
  labs(x="Date",y="Retweets",title="Tweeting by Trump") +
  scale_y_continuous(label=scales::comma)
```

Looking with `plotly`

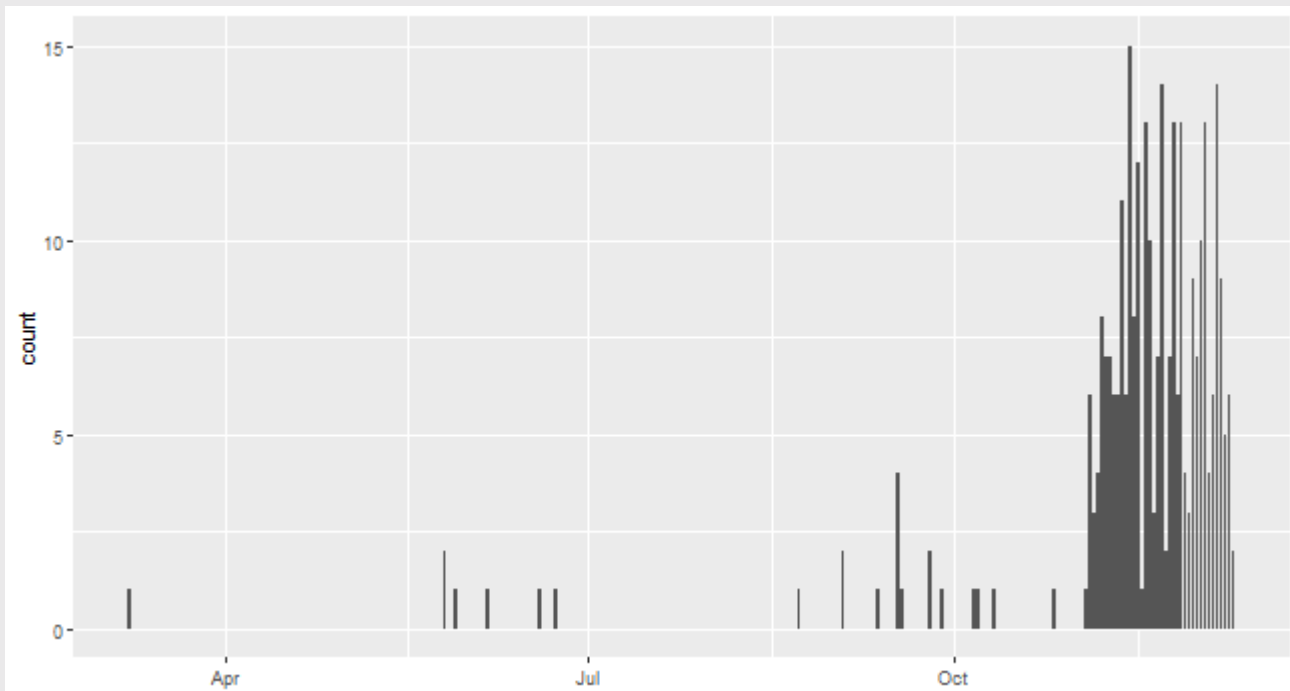
```
ggplotly(gg, tooltip = "text")
```



Trump's Censored Tweets

- What was Trump tweeting about when he was flagged by Twitter?

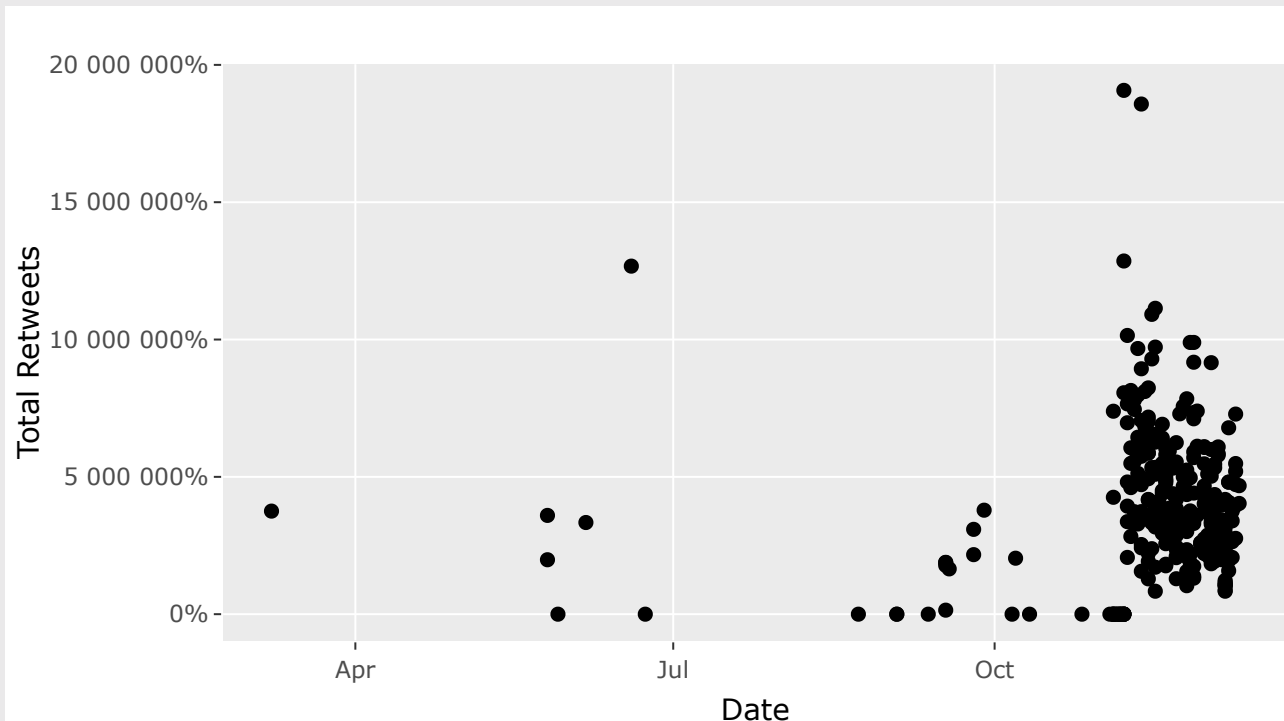
```
tweets %>%  
  filter(is_flagged) %>% # Indicator for Twitter flag  
  ggplot(aes(x = Tweeting.date)) +  
  geom_bar()
```



Looking at content

- We could just look at the flagged tweets themselves

```
ggplotly(gg)
```



Looking at content

- Or we can look at statistical summaries of Trump's tweets
 1. Tweets as **data**: we can understand him better
 2. Tweets as **object of interest**: how do his tweets influence public discourse?
- We will be using pre-processed tweets
 - Lots of **data wrangling** went into this
 - See the `prepping_raw_tweets.R` file for more details

```
tweet_words <-  
read_rds(file="https://github.com/jbisbee1/DS1000_F2024/raw/main/data/T
```

NLP Definitions

- Before we dig in, some important definitions

1. **Word / Term:** The core unit of interest

- Often pre-processed to remove "stop words" and to "stem" the words
- "Stop word": an uninteresting, commonly used word
- "Stem / Lemmatize": the core component of a word that contains its meaning (eat, ate, eaten → eat)

2. **Document:** A collection of words with a single purpose / idea (i.e., a tweet, an essay)

3. **Corpus:** A collection of documents

4. **BOW:** Bag-of-words. Convert a **document** into a count of how many times **words** appear

5. **DTM:** Document-term matrix. A dataset where rows are **documents**, columns are **words**, and the values are the counts from **BOW**

What does Trump tweet about?

- Core idea: word frequencies can help:
 - Help us understand **documents**
 - Which help us understand **authors**

```
counts <- tweet_words %>%  
  count(word) %>%  
  arrange(-n)
```


What does Trump tweet about?

counts

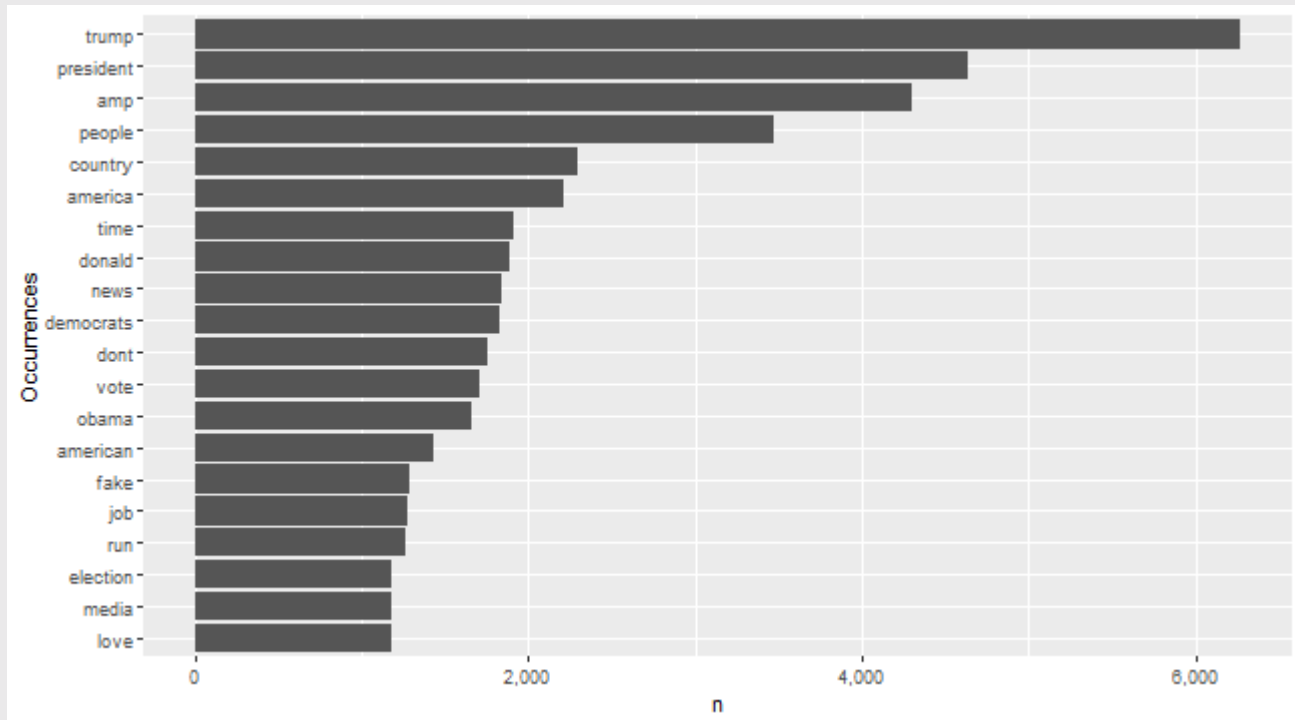
```
## # A tibble: 45,221 × 2
##   word          n
##   <chr>      <int>
## 1 trump      6269
## 2 president  4637
## 3 amp        4306
## 4 people     3475
## 5 country    2302
## 6 america    2211
## 7 time       1913
## 8 donald     1891
## 9 news       1842
## 10 democrats 1824
## # i 45,211 more rows
```

What does Trump tweet about?

```
p <- tweet_words %>%  
  count(word, sort = TRUE) %>% # New ways of doing old things  
  head(20) %>%  
  ggplot(aes(x = n, y = reorder(word, n))) +  
  geom_bar(stat = "identity") +  
  ylab("Occurrences") +  
  scale_x_continuous(label=comma)
```

What does Trump tweet about?

p



Effect of becoming president

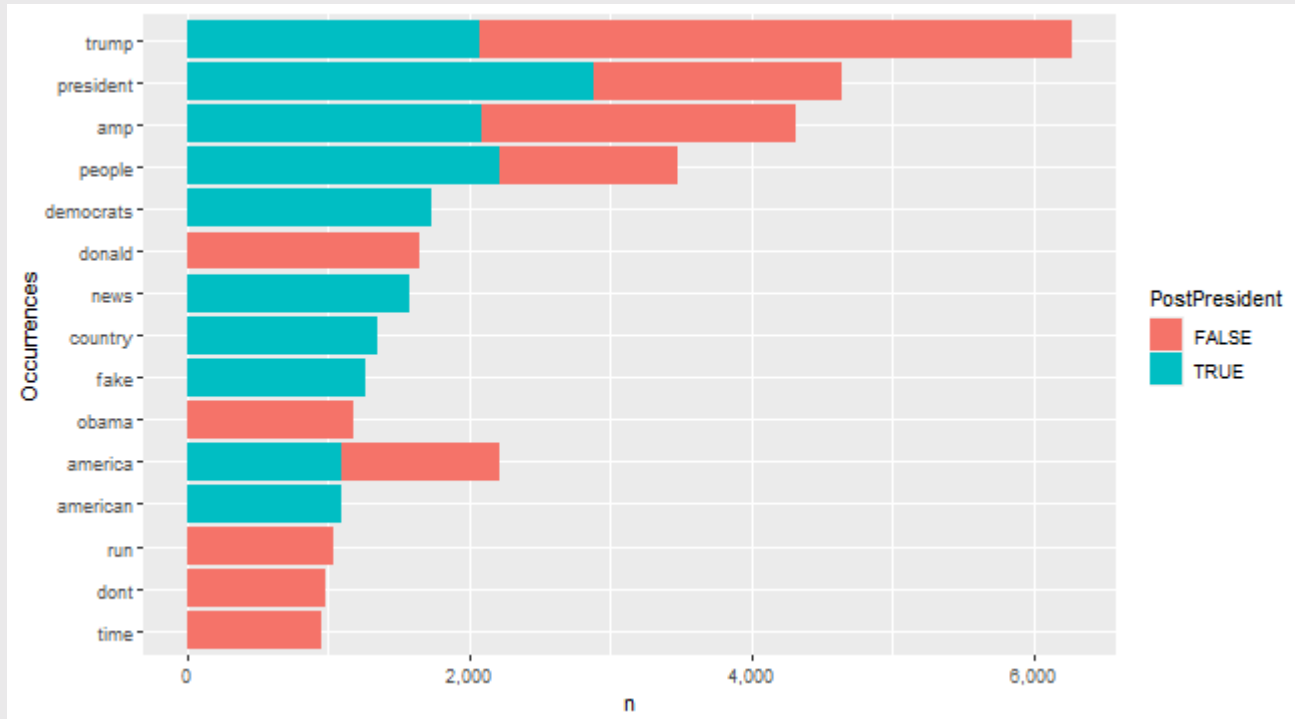
- Did his focus change when he became president?

```
tweet_words <- tweet_words %>%  
  mutate(PostPresident = Tweeting.date > as.Date("2016-11-03"))
```

```
p <- tweet_words %>%  
  count(PostPresident, word) %>%  
  group_by(PostPresident) %>%  
  arrange(-n) %>%  
  slice(1:10) %>%  
  ggplot(aes(x = n, y = reorder(word, n),  
            fill = PostPresident)) +  
  geom_bar(stat = "identity") +  
  ylab("Occurrences") +  
  scale_x_continuous(label=comma)
```

Effect of becoming president

p



Document Term Matrix

- "DTM" counts all the words in each document
- In this case, a "document" is a tweet

```
dtm <- tweet_words %>%  
  count(document,word)  
glimpse(dtm)
```

```
## Rows: 433,271  
## Columns: 3  
## $ document <dbl> 1698308935, 1698308935, 1698308935, 16983...  
## $ word      <chr> "david", "donald", "late", "letterman", "...  
## $ n         <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
```

Document Term Matrix

- However, each tweet is very short
- Let's consider the `Tweeting.date` the document
 - Concept: What is Trump tweeting about on a given day?

```
dtm <- tweet_words %>%  
  count(Tweeting.date,word) %>%  
  group_by(word) %>%  
  mutate(tot_n = sum(n)) %>% # Also calculate TOTAL number of times a  
word appears  
  ungroup()
```

Wrangle

- Extremely rare words should be dropped (typos, etc.)

```
dtm %>%  
  arrange(tot_n) %>% head() # Trump tweeted "barnesandnoblecom" only  
  once in his life
```

```
## # A tibble: 6 × 4  
##   Tweeting.date word                n tot_n  
##   <date>         <chr>              <int> <int>  
## 1 2009-05-08     httptinyurlcomooafwn      1      1  
## 2 2009-05-08     httptinyurlcomqluxe      1      1  
## 3 2009-05-12     cling                    1      1  
## 4 2009-05-12     wallflower               1      1  
## 5 2009-05-13     httptinyurlcomqsvl       1      1  
## 6 2009-05-15     contesthttpwwwtrumpthinklikeeach... 1      1
```

```
dtm <- dtm %>%  
  filter(tot_n > 20) # Drop these rarely occurring words
```


Analyzing BOW

- Some words are frequently found in many documents
- We want to find words that are **uniquely** used
 - "Unique" → used frequently in one document but not in any others
- "TF-IDF": Term frequency-inverse document frequency
- "TF": $\frac{\text{word count}}{\text{total words}}$
- "DF": $\frac{\text{documents with word}}{\text{total documents}}$
 - "IDF": Just invert it $\frac{\text{total documents}}{\text{documents with word}}$

$$tf - idf(w, d) = tf(w, d) \times \log \left(\frac{N}{df(w)} \right)$$

TF-IDF

```
require(tidytext) # Required to calculate TF-IDF
dtm.tfidf <- bind_tf_idf(tbl = dtm, term = word, document =
  Tweeting.date, n = n) # Calculate TF-IDF
dtm.tfidf %>%
  select(word,tf_idf) %>%
  distinct() %>%
  arrange(-tf_idf) %>%
  slice(1:10)
```

```
## # A tibble: 10 × 2
##   word      tf_idf
##   <chr>      <dbl>
## 1 gma        5.21
## 2 generation 4.90
## 3 dead       3.19
## 4 apprentice 2.58
## 5 weekly     2.40
## 6 appearance 2.26
## 7 answers    2.11
## 8 sounds     2.10
## 9 football   2.01
## 10 defeat    1.90
```

K-means

- How to summarize this? *k*-means clustering!
- Recall that `kmeans()` function clusters over every column in a data frame
- `dtm.tfidf` is organized "long" (i.e., each row is a word-by-document)
- Want to convert to "wide" (i.e., rows are documents, columns are words)
- Previously used `spread()`, but for *k*-means with text: `cast_dtm()`

```
castdtm <- cast_dtm(data = dtm.tfidf, document = Tweeting.date, term  
= word, value = tf_idf)
```

- Now let's calculate `kmeans()` (this will take a few seconds)

```
set.seed(42)  
km_out <- kmeans(castdtm,  
                 centers = 50, # Number of "topics"  
                 nstart = 5)
```

Looking at Clusters

- Some quick wrangling

```
require(tidymodels)
km_out_tidy <- tidy(km_out) %>%
  gather(word, mean_tfidf, -size, -cluster, -withinss) %>% # Convert to
  Long data
  mutate(mean_tfidf = as.numeric(mean_tfidf)) # Calculate average TF-
  IDF
km_out_tidy
```

```
## # A tibble: 158,850 × 5
##   size withinss cluster word mean_tfidf
##   <int>   <dbl> <fct>   <chr>   <dbl>
## 1     12    12.6     1    david     0
## 2      1      0      2    david     0
## 3      1      0      3    david     0
## 4      4     7.12     4    david     0
## 5     16    12.4     5    david     0
## 6     36    43.0     6    david     0
## 7    156    54.5     7    david 0.000647
## 8      1      0      8    david     0
## 9      1      0      9    david     0
```

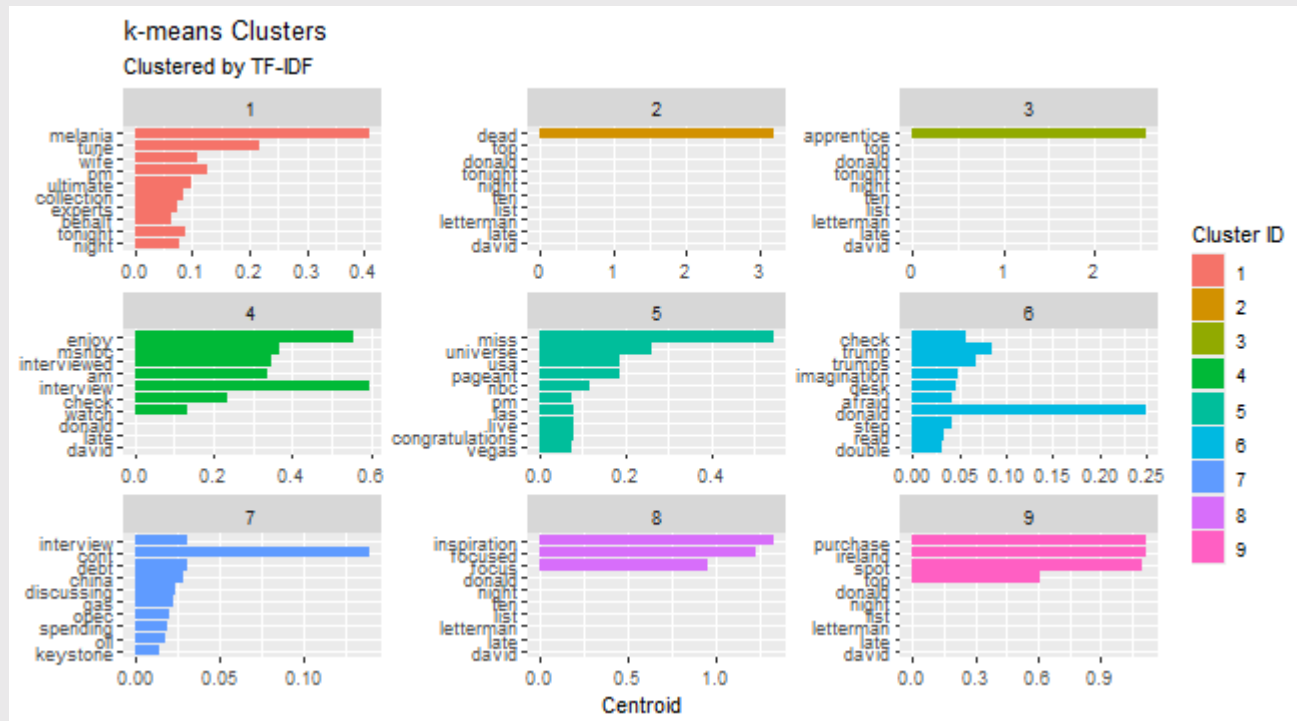
Looking at Clusters

- And can plot! (Just look at first 10 "topics")

```
p <- km_out_tidy %>%  
  filter(cluster %in% 1:9) %>%  
  group_by(cluster) %>%  
  arrange(-mean_tfidf) %>%  
  slice(1:10) %>%  
  ggplot(aes(x = mean_tfidf, y = reorder(word, mean_tfidf),  
            fill = factor(cluster))) +  
  geom_bar(stat = 'identity') +  
  facet_wrap(~cluster, scales = 'free') +  
  labs(title = 'k-means Clusters',  
       subtitle = 'Clustered by TF-IDF',  
       x = 'Centroid',  
       y = NULL,  
       fill = 'Cluster ID')
```

Looking at Clusters

p



Looking at clusters

- What are these clusters?
- Let's look at the most popular topics

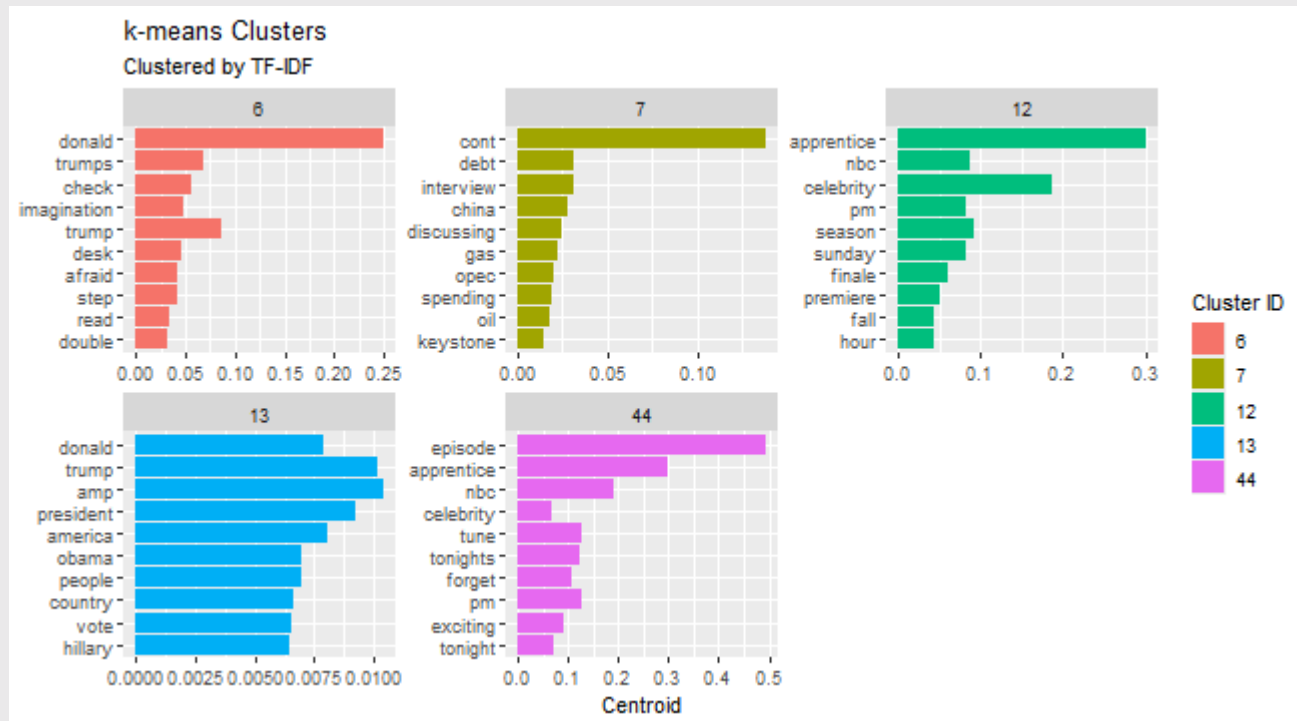
```
(tops <- km_out_tidy %>%  
  select(size,withinss,cluster) %>%  
  distinct() %>%  
  arrange(desc(size)) %>%  
  slice(1:5))
```

```
## # A tibble: 5 × 3  
##   size withinss cluster  
##   <int>    <dbl> <fct>  
## 1  3066    720.    13  
## 2   156    54.5     7  
## 3    36    43.0     6  
## 4    34    28.1    12  
## 5    17    13.9    44
```

```
p <- km_out_tidy %>%  
  filter(cluster %in% tops$cluster) %>%  
  group_by(cluster) %>%
```

Looking at clusters

p



Optimal number of clusters?

- We know how to do this already! Elbow plot!

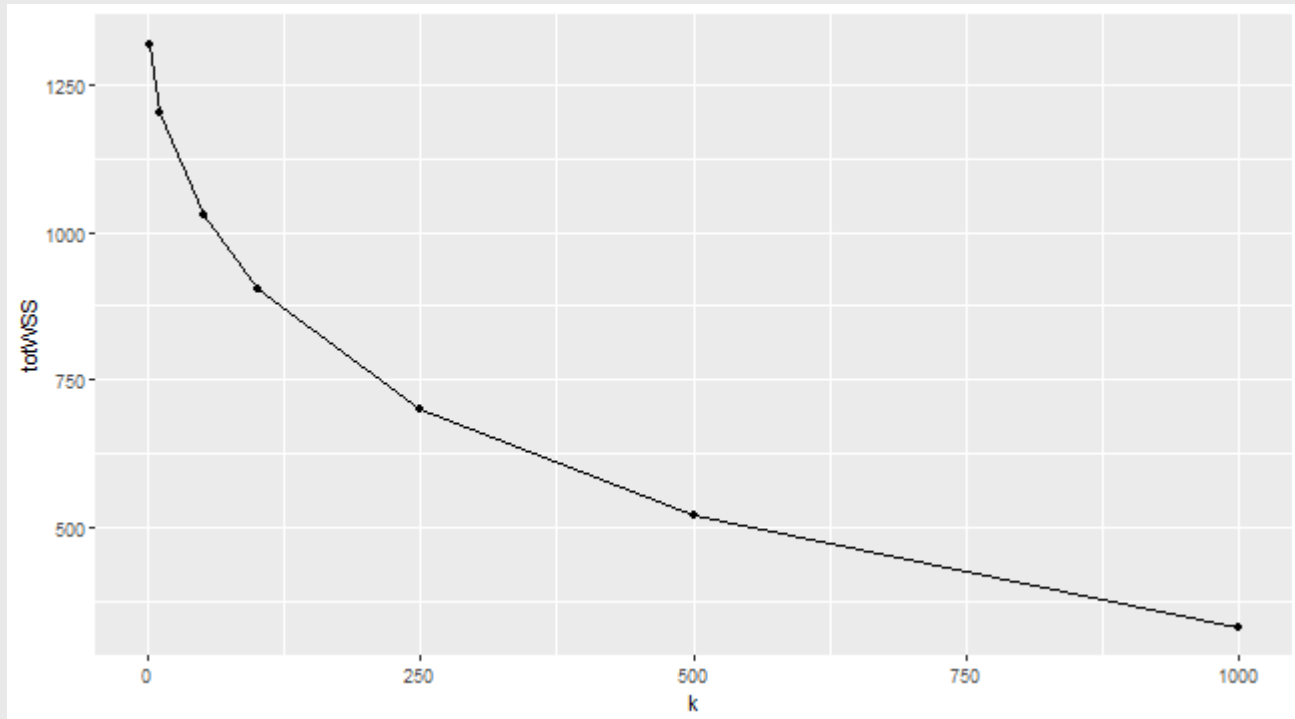
```
set.seed(42)
totWSS <- NULL
for(k in c(1,10,50,100,250,500,1000)) {
  km_out <- kmeans(castdtm,
                  centers = k,
                  nstart = 5)

  totWSS <- data.frame(totWSS = km_out$tot.withinss,
                      k = k) %>%
    bind_rows(totWSS)
}

p <- totWSS %>%
  ggplot(aes(x = k, y = totWSS)) +
  geom_point() +
  geom_line()
```

Optimal number of clusters?

p



Conclusion

- k -means clustering on text → **topics**
- As always, this is a deep area of study
 - Superior methods are out there (Latent Dirichlet Allocation, Structural Topic Modeling, etc.)
- NOTE: even with text, always start with simple descriptives
 - **Looking** at your data is the heart of data science

